

# AST 201

## Stars & Galaxies

ast201@astro.utoronto.ca

Credit: NASA, ESA, CSA, STScI, Webb ERO Production Team

# Recap: Two Models of Gravity

- Newtonian:
  - gravity is a force that is transmitted between objects through flat space
  - gravity travels instantaneously
- Einsteinian:
  - objects try to follow the straightest possible path through spacetime that is curved by the mass and energy of other objects
  - changes in spacetime curvature move at the speed of light

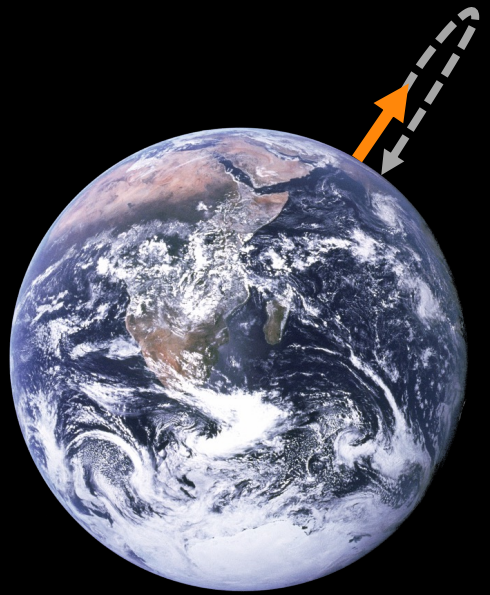
# Introduction to Black Holes



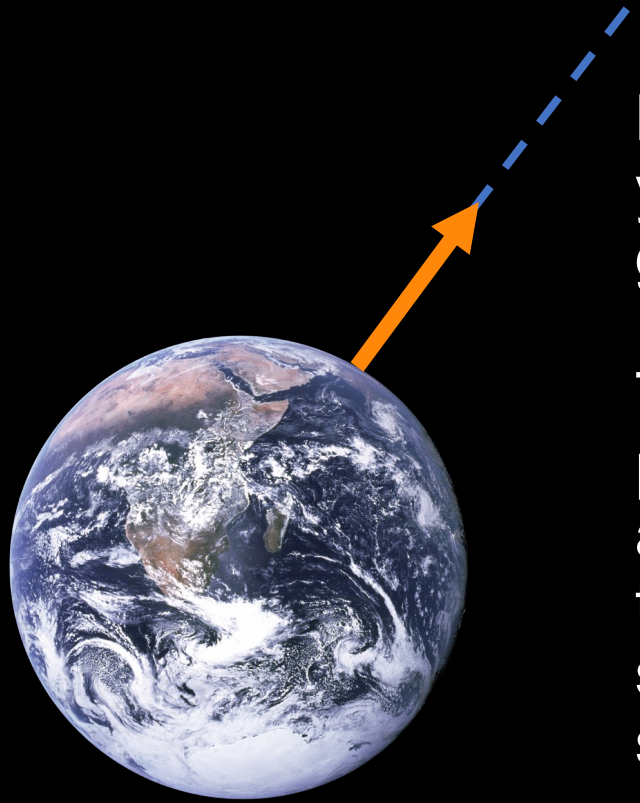
What comes to mind when you think of black holes?



To understand black holes, think about trying to escape the gravity of Earth.



Launch too slow and
you fall back to Earth.

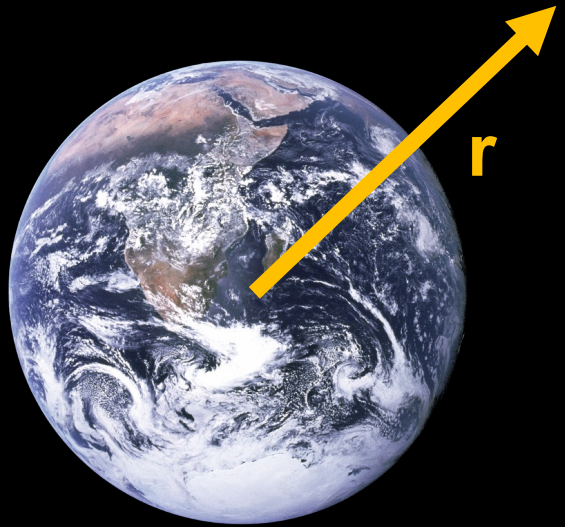


Launch fast enough, and you break free from Earth's gravity.

The speed required for a projectile to break free from an object's gravity is called the **escape speed**. From the surface of Earth, the escape speed is about 11.2 km/s.*

*Note that the concept of escape speed only applies to unpowered projectiles, such as a cannon ball. The situation is different if the object has an engine an can accelerate continuously, like a rocket.

The **escape speed** depends on the mass of the object you're trying to escape from, M, and your distance, r, from from the center of mass:




$$v_{escape} = \sqrt{\frac{2GM}{r}}$$



What would happen to the escape speed from the surface of Earth if we could compress the planet?

Let’s explore...



What would happen if we could compress Earth so much that the escape speed **reached the speed of light**?



In that case, nothing would be able to leave Earth—not even light.

Earth would become a **black hole**.



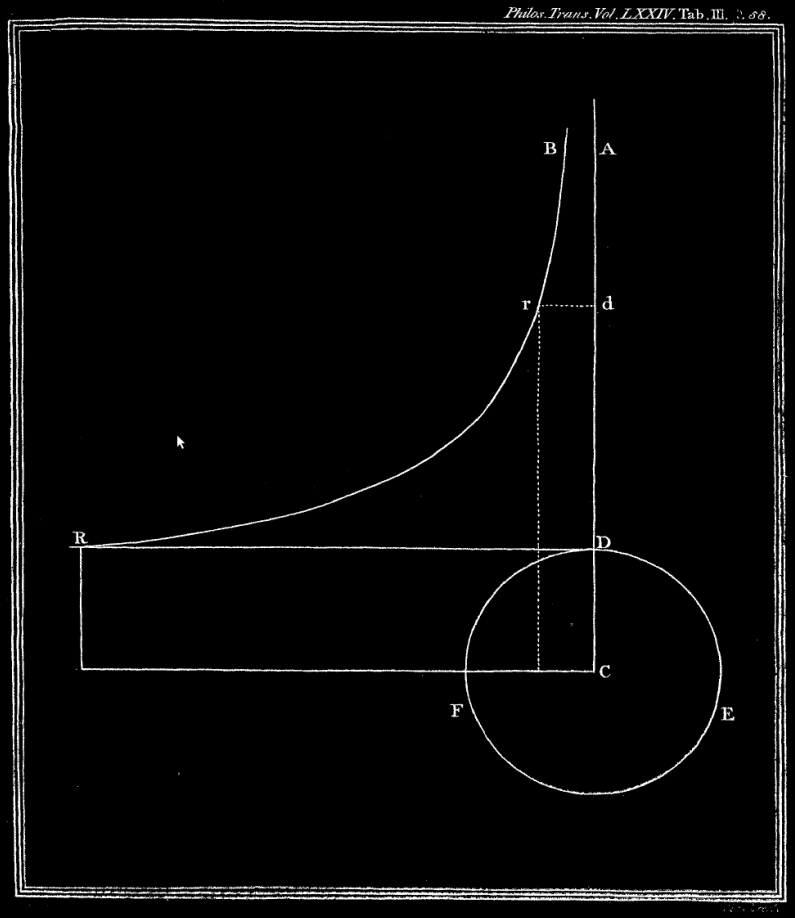

A black hole is a region of space where matter is compacted so densely that you'd have to move faster than the speed of light to escape its gravity.




The outer boundary of a black hole is called the **event horizon**. This is the distance from the center at which the escape speed equals the speed of light.

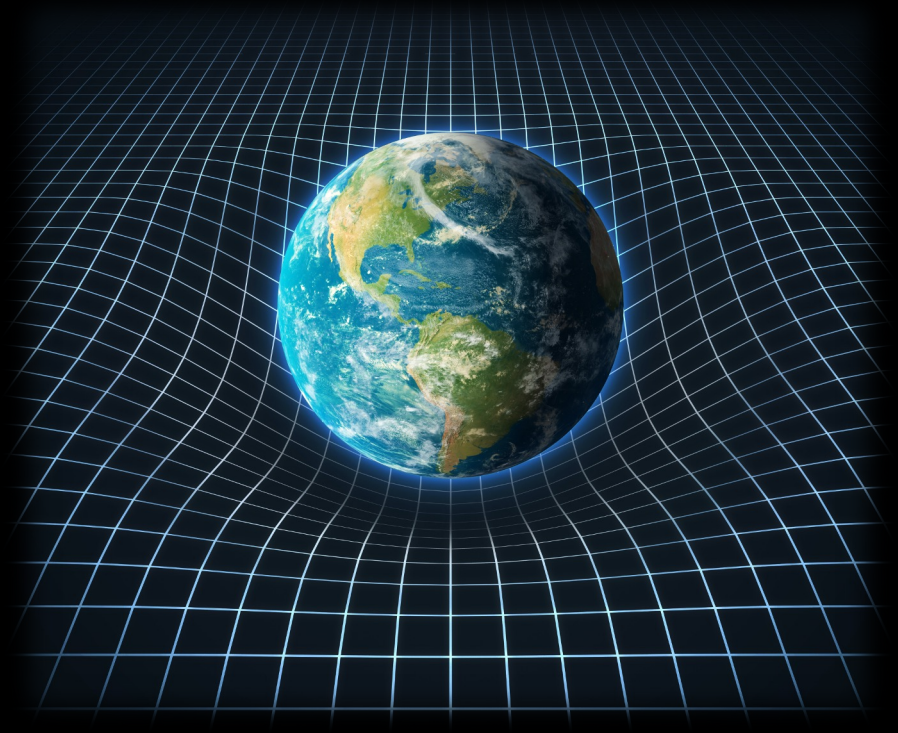



The **singularity** at the center of the black hole is where the density goes essentially to infinity. We don't have the proper physics to describe this yet.




The essential idea of a black hole has been around for centuries. It dates back at least as far as John Michell (1724-1793), who proposed “dark stars” so massive that their own light could not escape their gravity.

This version of a black hole is based on **Newtonian gravity**.



Remember that Einstein's **general relativity** replaces the Newtonian concept of gravity as a standard force.

Instead, it describes how mass curves spacetime, forcing objects to follow the straightest possible trajectory through that curved spacetime.

Credit: Adobe Stock



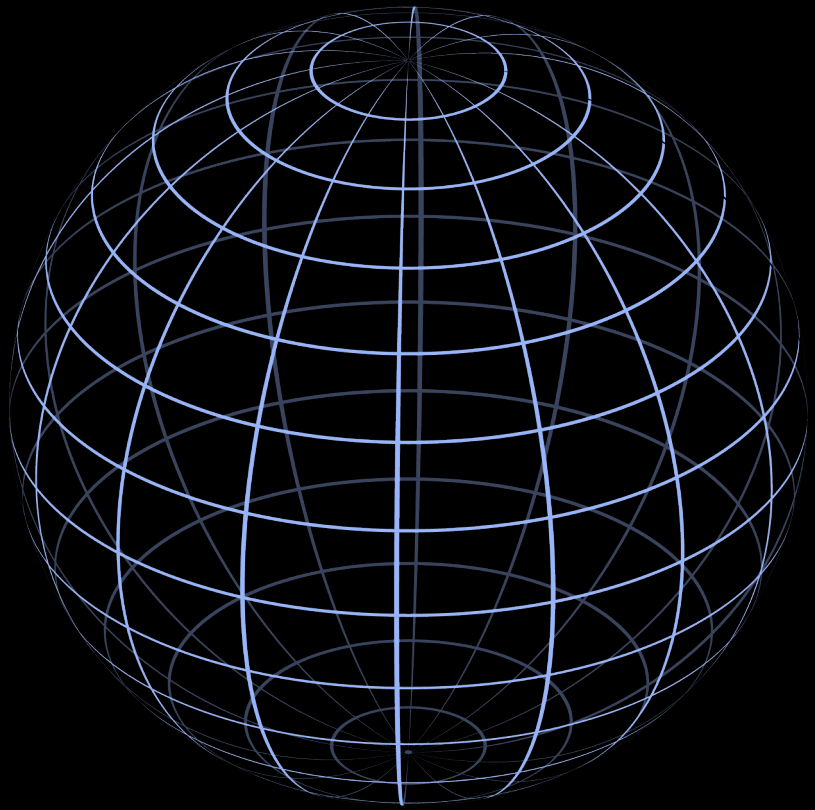
Copyright © 2008 Pearson Education, Inc., publishing as Pearson Addison-Wesley.

According to general relativity, a black hole is a region of spacetime bent so severely you would need to move faster than the speed of light to escape. In one sense, you can think of spacetime as “flowing” into the black hole at the speed of light.




Remember that while we sometimes represent black holes as “dents” in two-dimensional sheets, this is only for the convenience of representing them on two-dimensional screens.

**Black holes are *not* shaped like holes in the ground.**



event horizon

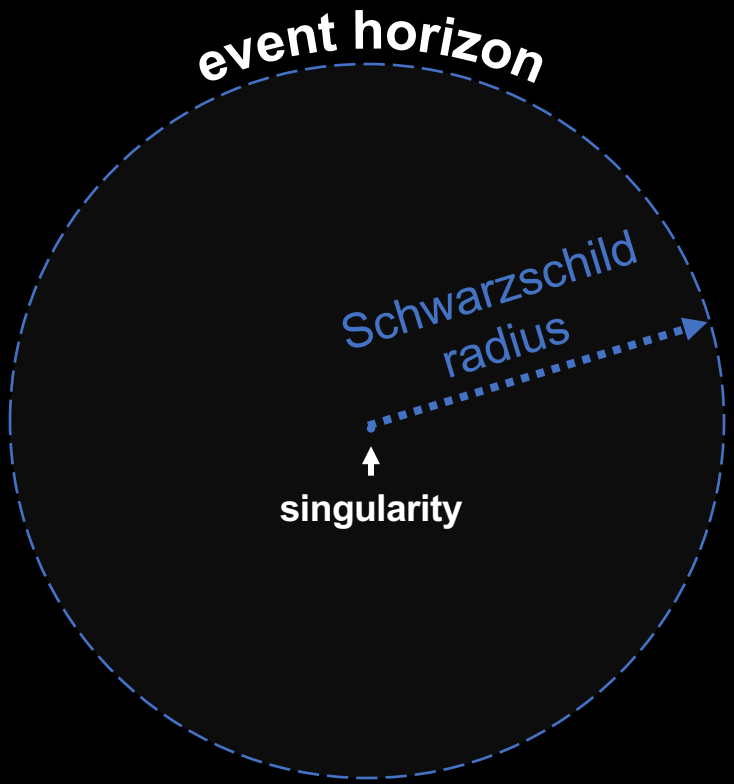
Black holes are fundamentally three-dimensional objects. The event horizon of a simple*, black hole would be a sphere.

*A non-rotating black hole with no electric charge. We won't cover those more complicated kinds of black holes in this course.

The event horizon is not a physical structure. It's analogous to the border between countries: an imaginary line that has very real consequences.

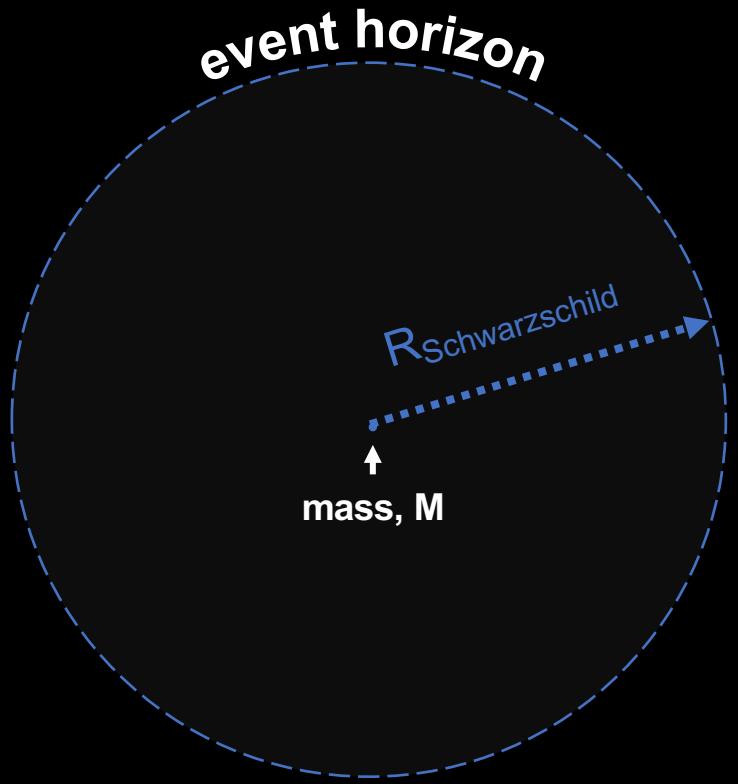
The physical object from which the black hole formed, or whatever remains of it, is buried deep inside, perhaps becoming the singularity.*

*Maybe? We don't really know.**
**I debated this with my colleagues on Slack, and none of us are at all sure how this works. It is very much up for debate. And we probably just don't know yet. Maybe YOU will figure it out!




The distance from the singularity to the event horizon is called the **Schwarzschild radius**.

The **Schwarzschild radius** depends *only* on the mass of the black hole:




$$R_{Schwarzschild} = \frac{2GM}{c^2}$$

Loosely, you can think of the Schwarzschild radius as the size you would have to compress an object to for it to form a black hole.

# Schwarzschild radii of some objects:

The Milky Way Galaxy ~ 0.3 light years

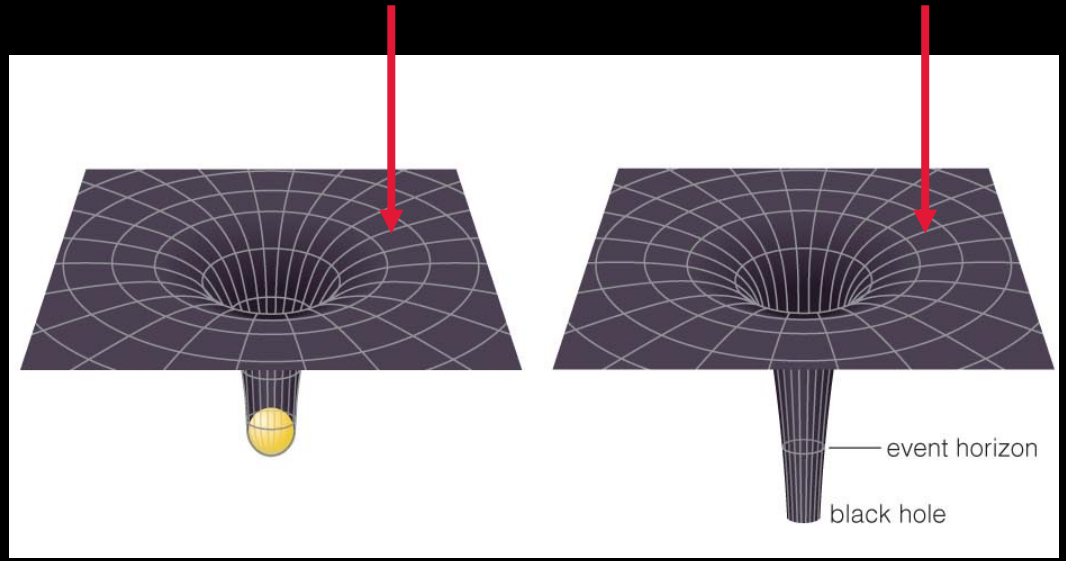
The Sun ~ 3 km

Jupiter ~ 3 m

Earth ~ 1 cm

a strawberry ~ $10^{-29}$ m

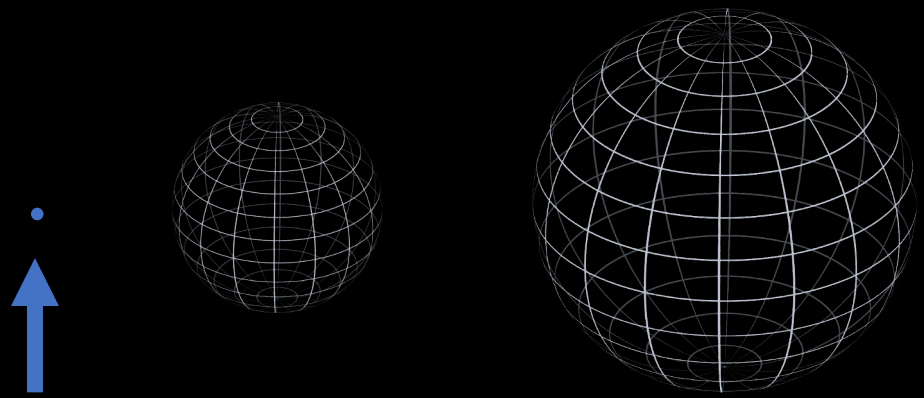
The spacetime curvature (gravity) at these points is the same, as long as the mass of the central object is the same.



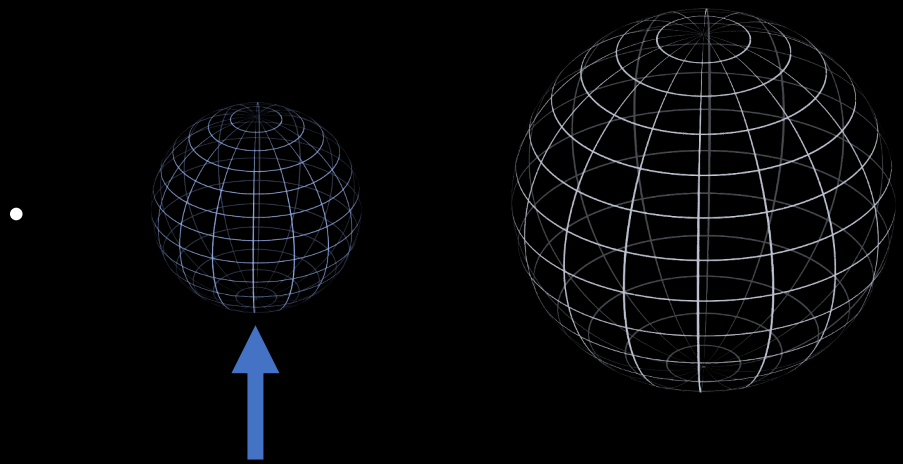

Because black holes have this unusual one-way outer boundary (the event horizon), people often imagine that black holes somehow acquire “extra” gravity.

This is NOT true. At a distance, a black hole has exactly as much gravity (or spacetime curvature) as whatever it formed from.

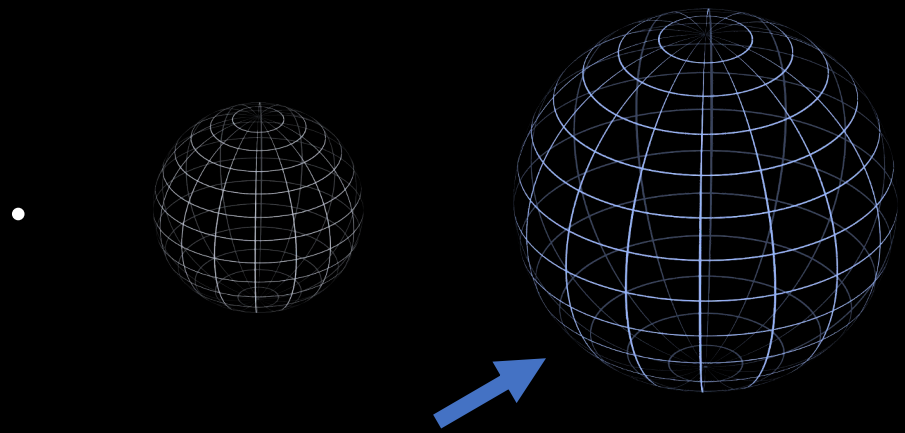
There are different ways to classify black holes, often based on either their formation mechanism or their mass. Those two things tend to be related.



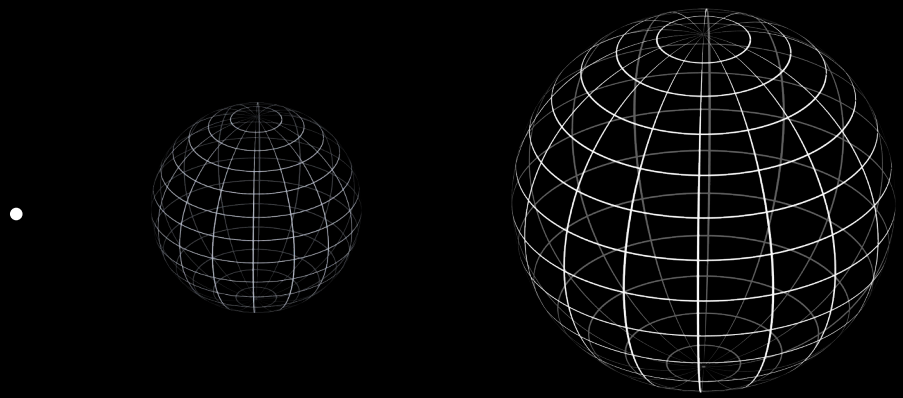
**Microscopic black holes** are hypothetical objects that could be produced during particularly energetic particle collisions. We have no evidence that they exist.



**Stellar mass black holes** have masses in the range of around 5-100 solar masses. They form when very massive stars explode as supernovas. There are likely hundreds of millions of these in our Milky Way galaxy. We have identified several dozen.



**Intermediate mass black holes** have masses in the range from around 100 to 100,000 solar masses. These may form from the collision of smaller objects, or they may have originated in the early universe. The evidence that they exist comes mainly from gravitational waves.



**Supermassive black holes** often have masses of millions or billions of solar masses. They reside at the hearts of most large galaxies. We are still trying to work out how they formed. They may have formed by direct collapse, or as stellar mass black holes that grew.

What would you see if you fell into a black hole?

Let’s look at a simulated view of falling into a supermassive black hole with a mass of about 5 million solar masses—roughly the same as the black hole at the center of our galaxy.



Credit: Andrew Hamilton, University of Colorado

For a sufficiently large black hole, you could survive passage through the event horizon (before dying a few seconds later), but you wouldn't survive passage into a stellar-mass black hole because of its more extreme **tidal forces**.

A **tidal force** is not a new force, but rather a difference in the force of gravity* between two points.

*Yes, I am using the language of “forces of gravity” again, even though I should be talking about spacetime curvature. We get pretty lazy about this.



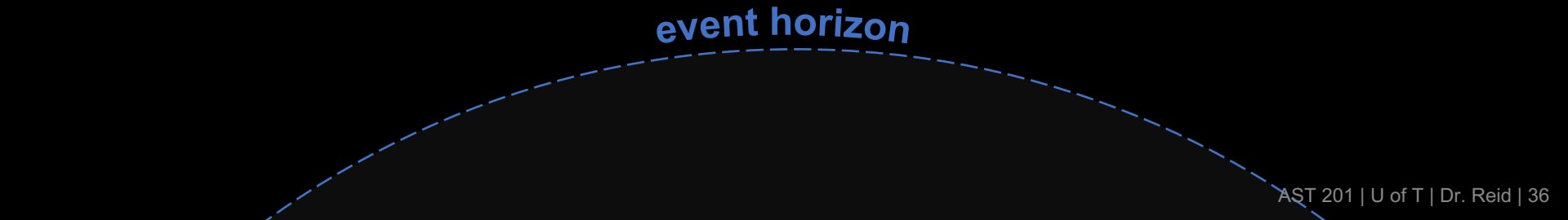

When you are near the surface of Earth, you feel *very slightly* more gravity on your feet than your head.

This tiny tidal force is too small to notice.




As you fall toward a black hole, the spacetime curvature is so extreme that the force of gravity on your feet grows MUCH larger than that on your head.

We say that the tidal force (the difference in the force) is very large.



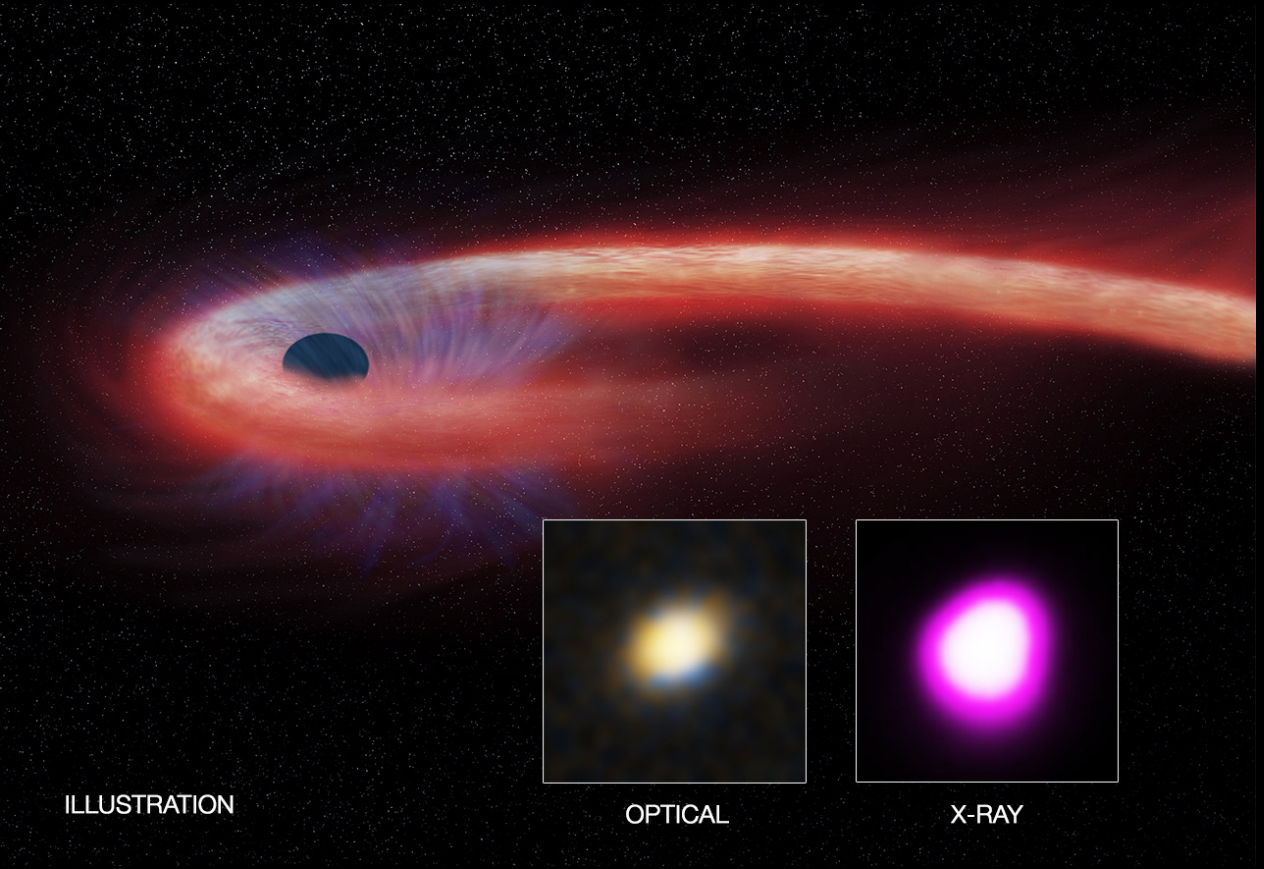




event horizon

Objects falling into black holes get “spaghettified” by these differences in the force of gravity across their bodies.


These tidal forces can lead to “tidal disruption events” of anything falling into black holes—including entire stars!
Credit: DESY, Science Communication Lab




Using X-ray telescopes, we can sometimes detect these tidal disruption events.

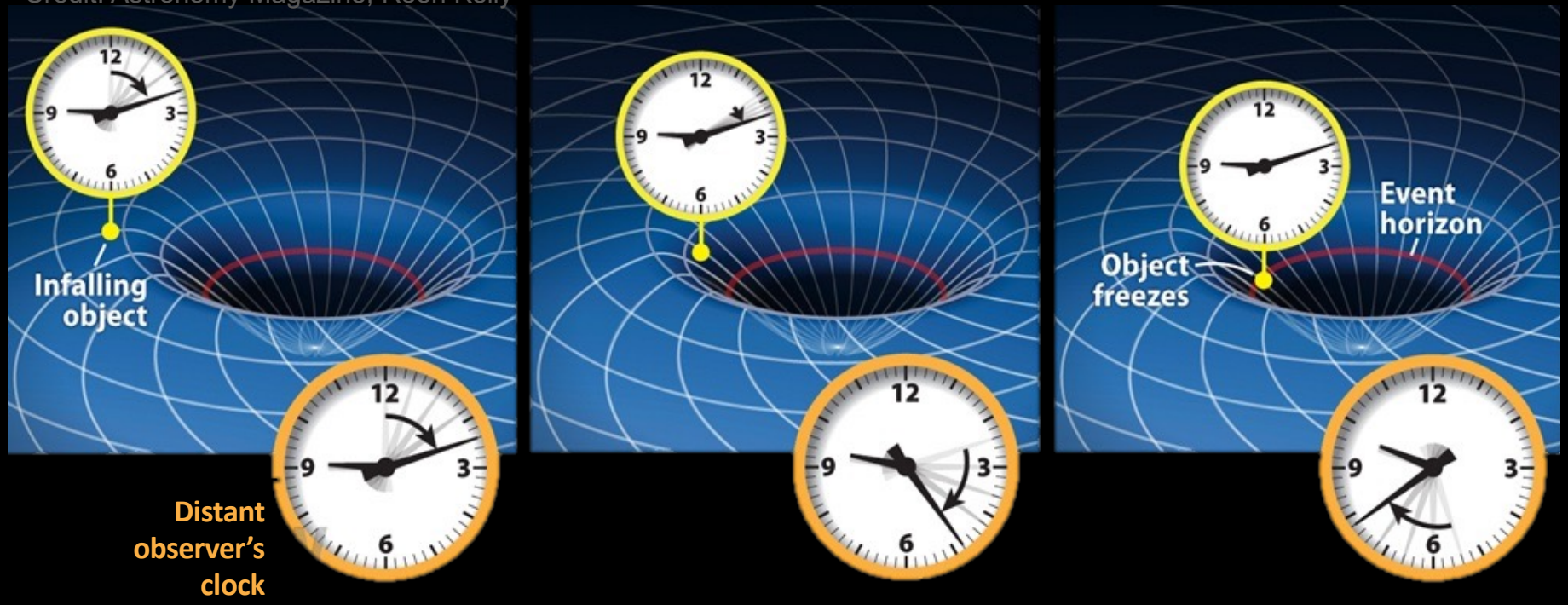
Huge flares of high-energy X-rays are emitted as stars and other objects are shredded upon close approach to a black hole.

Credit: X-ray: NASA/CXC/UNH/D.Lin et al, Optical: CFHT, Illustration: NASA/CXC/M.Weiss

Remember that relativity binds space and time together into a single entity, called spacetime. Mass deforms spacetime.

This gives rise to the phenomenon of **gravitational time dilation**, which is extreme in the case of black holes.

Credit: Astronomy Magazine, Roen Kelly




From the perspective of an observer distant from a black hole, a clock falling into a black hole would appear to tick slower and slower until it ultimately stopped as it crossed the event horizon.